Aravinth Siva Subramaniam Ekamparam, Ph.D.

Environmental Engineer | *Early Career Researcher*

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RESEARCH PROFILE

Environmental geochemist with 10+ years of experience in mineral–water interfacial processes, geochemical characterisation of soils and sediments, contaminant fate and transport, and physicochemical water/wastewater treatment. PhD from IIT Kanpur and three years of postdoctoral research at Oregon State University (USA), with hands-on expertise in ICP-MS/OES, XRD, XPS, HRTEM-SAED, and column experiments. Current research focuses on soil and sediment geochemical characterisation, contaminant speciation and mobility, and physicochemical treatment of complex water matrices.

KEY COMPETENCIES

- Geochemical characterisation of soils, sediments, and contaminated materials (XRD, XPS, HRTEM-SAED, FTIR, ICP-MS, grain size analysis)
- Pretreatment development for managed aquifer recharge systems: porous media filtration, clogging assessment, and backflushing strategies
- Column experiments and batch studies for contaminant transport, clogging dynamics, and water quality evaluation
- Groundwater quality monitoring: physicochemical analysis, surface water intrusion assessment, and subsurface geochemistry
- Field instrumentation and sensor-based monitoring: pressure, turbidity, conductivity, and water level data integration
- Experimental design, multi-element analytical data interpretation, and decision-support framework development for water supply systems

RESEARCH AREAS

Environmental Geochemistry	Groundwater Quality and Water Safety
Managed Aquifer Recharge	Physicochemical Water Treatment

EDUCATION

Doctor of Philosophy in Environmental Engineering 2014–2021
Indian Institute of Technology Kanpur, Kanpur, India
Dissertation Advisor: Dr. Abhas Singh; *Dissertation Title:* Investigation of processes governing fluoride removal from water by calcium-based amendments.

Master of Engineering in Environmental Engineering 2012–2014
Anna University, Chennai, India
Thesis Advisor: Dr. S. Pauline; *Thesis Title:* Modeling of transport and biotransformation of As(V) in confined aquifers.

Bachelor of Engineering in Civil Engineering 2008–2012
Anna University, Chennai, India
Project Advisor: Dr. D. Daniel Thangaraj; *Project Title:* Ductile design of earthquake-resistant building; *Mini-Project Title:* Analysis and design of earthquake-resistant multi-storied building.

EMPLOYMENT

Research Scientist May 2026–Present
Detoxyfi Technologies Pvt. Ltd., India

Postdoctoral Scholar Oct 2022–Oct 2025
Oregon State University, USA
Supervisor: Dr. Salini Sasidharan

Project Manager Jul 2022–Sep 2022
Indian Institute of Technology Madras, India
Supervisor: Dr. Ligy Philip

Temporary Faculty Aug 2021–May 2022
National Institute of Technology Tiruchirappalli, India

Senior Project Scientist Feb 2021–Aug 2021
Indian Institute of Technology Kanpur, India
Supervisor: Dr. Abhas Singh

WORK EXPERIENCE

Drywell Managed Aquifer Recharge 2023–2025

- Installed drywells, vadose zone wells, and integrated multiple sensors at the North Willamette Research & Extension Center (NWREC), Aurora, Oregon.
- *Result: Completed the first recharge trial using 10,000 gallons of water and initiated continuous water quality monitoring.*

Solar Panel Collected Water for Drywell Managed Aquifer Recharge 2024–2025

- Designed and implemented a rainwater harvesting system integrated with a solar-tracking agrivoltaics infrastructure at NWREC, with the system designed for diversion of harvested water to drywell managed aquifer recharge (MAR) systems.
- Collected and analyzed water samples for multiple parameters including non-targeted analysis.
- *Result: Demonstrated potential to utilize harvested water for groundwater recharge via drywells.*

Development of Pretreatment Unit for Drywell Managed Aquifer Recharge Systems 2022–2025

- Conducted physical clogging experiments using a sensor-fitted column under constant head conditions with turbidity levels ranging from 250–7000 FNU.
- Evaluated unclogging strategies such as optimized backflushing and sacrificial sand replacement.
- *Result: Identified porous media with potential as effective pretreatment units to extend drywell MAR lifespan*

Assessment of In situ Clogging of a Infiltration Basin 2024–2025

- Designed and installed a 6-ft infiltration column equipped with pressure sensors and flow meters.
- Packed with natural soil and sand layers to simulate infiltration system conditions and evaluated physical clogging under variable head.
- *Result: Determined infiltration capacity of native soil and proposed coarse sand layering to improve recharge efficiency.*

Natural Polymer based Treatment Units 2024–2025

- Coated quartz sand with natural polymers to enhance sorption of chromium, arsenic, and E. coli.
- Performed batch experiments to quantify sorption performance.
- *Result: Determined maximum sorption capacity and optimized treatment parameters.*

Bio-geochemistry of the nitrate-contaminated aquifer in Clackamas County, Oregon 2023–2025

- Performed leaching experiments on soil cores and collected groundwater samples for over a year.
- Conducted geochemical analyses of elemental and nutrient concentrations.
- *Result: Identified Mn and Fe leaching mechanisms under nitrate-rich conditions; ongoing study to quantify element mobilization.*

Point-of-Use Treatment Unit for Metal Polishing Industry Wastewater Jul 2022–Sep 2022

- Tested multiple treatment strategies for high-strength wastewater (COD 300,000 mg/L).
- *Result: Achieved 90% COD reduction in preliminary trials.*

Characterization of Chromium Ore Processing Residue (COPR) Soil Samples Feb 2021–Jul 2021

- Collected and analyzed Cr-contaminated soils.
- *Result: Quantified chromium concentrations across depth profiles to support remediation planning.*

Investigation of Calcium-Phosphate Systems for Fluoride Removal 2014-2021

- Performed batch and flow-through studies with calcite–phosphate systems; developed mass-balance and nucleation models.
- Applied XRD, HRTEM-SAED, and XPS to confirm mineral phases and surface complexation models to explain adsorption.
- *Results: Identified fluorapatite precipitation as the principal removal mechanism and quantified formation kinetics.*

Transport and Bio-transformation of As(V) in Confined Aquifers 2013-2014

- Developed a transport–reaction mass-balance model solved with the 4th-order Runge–Kutta method.
- Integrated growth kinetics of Bacillus arsenicus into the model.
- *Result: Elucidated coupled microbial and geochemical controls on As(V) fate.*

Structural Analysis & Design of Multi-storied Building 2011–2012

- Conducted seismic load analyses in ANSYS and STAAD Pro and validated with manual calculations.
- *Result: Delivered structural designs fully compliant with earthquake-resistant codes.*

PUBLICATIONS

- **Ekamparam, A. S. S.**; Bradford, S. A.; Sasidharan, S., Pretreatment Approaches to Minimize Clogging During Managed Aquifer Recharge with Drywells. Colloid. Surf. A. Phys. Eng. Asp. (2025) 726 137891.
- **Ekamparam, A. S. S.**; Sujathan, S.; Singh, A., Kinetics of Fluorapatite Precipitation in Fluoride-Contaminated Water. ACS ES&T Eng. (2024) 4 (10) 2370-2380.

- **Ekamparam, A. S. S.**; Khaitan, H.; Nimesh, V.; Singh, A., Relative Extents, Mechanisms, and Kinetics of Fluoride Removal from Synthetic Groundwater by Prevalent Sorbents. Chemosphere (2023) 342 140161.
- Mohapatra, A. K.; Sujathan, S.; **Ekamparam, A. S. S.**; Singh, A., The Role of Manganese Carbonate Precipitation in Controlling Fluoride and Uranium Mobilization in Groundwater. ACS Earth Space Chem. (2021) 5 2700-2714.
- **Ekamparam, A. S. S.**; Singh, A., Transformation of calcite to Fluorapatite at Room Temperature: Impact of Initial Phosphate and Fluoride Concentrations. Geochim. Cosmochim. Acta. (2020) 288 16-35.
- Pokhara, P.; **Ekamparam, A. S. S.**‡; Gupta, A. B.; Rai, D. C.; Singh, A., Activated Alumina Sludge as Partial Substitute for Fine Aggregates in Brick Making. Constr. Build. Mater. (2019) 221 244-252; ‡*Joint first author*.
- **Ekamparam, A. S. S.**; Muzhumathi, E. P., Degradation of Dyeing Effluent using Fenton's Oxidation. Res. J. Eng. Technol. (2013) 4(4) 217-220.

MANUSCRIPTS UNDER REVIEW

- Kumar, V.; Ranjan, S.; Saini, R. K.; **Ekamparam, A. S. S.**; Singh, R.; Sharma, M. K.; Tyagi, I.; Chandniha, S. K.; Kumar, T; Singh, S.; Krishan, G.; Kumar, S.; Yadav, B. K., Mineralogical and Geochemical Characterization of Sediments in Perspective of Arsenic Source and Mobilization in Shallow Alluvial Aquifers of Central Ganga Basin, India.

CONFERENCE PRESENTATIONS

- Gomes, R., **Ekamparam, A. S. S.**, Oti,J.,Namyasenko, G., Osterman, G., and Sasidharan, S. "Deep Vadose Sense: Bridging in-Situ Sensing, Geophysics, and Modeling for Sustainable Aquifer Recharge", SSSA Khirkham Conference, Fukushima, Japan (19-22 Aug 2025). [Poster]
- **Ekamparam, A. S. S.**, Bradford, S. A., and Sasidharan, S. "Effective Pre-treatment Approaches for Managed Aquifer Recharge: Preventing Clogging in Drywells", AGU Fall Meetings, Washington D.C., USA (9-13 Dec 2024). [Poster]
- Sasidharan, S., Bradford, S. A., **Ekamparam, A. S. S.**, Kumar, S., and Silva, T.T.M.E. "Assessing Drywell Efficiency in Agricultural Managed Aquifer Recharge (Ag-MAR) in Fresno, California", AGU Fall Meetings, San Francisco, California, USA (11-15 Dec 2023). [Oral]
- **Ekamparam, A. S. S.** and Sasidharan, S. "Evaluating Urban Stormwater Treatment Systems for Effective Managed Aquifer Recharge: A Comprehensive Framework", 2nd Annual Postdoc Research Symposium, Oregon Health & Science University, Portland, OR, USA (27 Oct 2023). [Poster]
- Singh, A. **Ekamparam, A. S. S.** and Sujathan, S. "Kinetics of Transformation of Calcite to Fluorapatite Under Flow conditions", Goldschmidt-2021, Virtual (4-9 July 2021). [Oral]
- **Ekamparam, A. S. S.** and Singh, A. "Phosphate Induced Transformation of Calcite to Fluorapatite in the Presence of Fluoride", Goldschmidt-2018, Boston, US (12-17 Aug 2018). [Oral]
- **Ekamparam, A. S. S.**, Pokhara, P., and Singh, A. "Understanding the Mechanisms of Fluoride Mobilization in Indian Groundwaters", 7th International Ground Water Conference (IGWC-2017), New Delhi, India (11-13 Dec, 2017). [Oral]
- **Ekamparam, A. S. S.**, Khaitan, H. Nimesh, V. and Singh, A. "Understanding Fluoride Sorption on Activated Alumina in Synthetic Water Systems", 7th International Ground Water Conference (IGWC-2017), New Delhi, India (11-13 Dec, 2017). [Oral]
- Mohapatra, A. K., **Ekamparam, A. S. S.** Mobilization in Groundwater of Bansathi, Kanpur Nagar, India", 7th International Ground Water Conference (IGWC-2017), New Delhi, India (11-13 Dec, 2017). [Oral]

- **Ekamparam, A. S. S.**, and Singh, A. "Dissolution-precipitation Kinetics of Fluoride-bearing Minerals in Groundwater", IUPAC Symposium 7th International Symposium on Solubility Phenomena and Related Equilibrium Processes (ISSP17), Geneva, Switzerland (24-29 July 2016). [Oral]
- **Ekamparam, A. S. S.**, and Singh, A. "Mechanisms of Fluoride Mobilization in Indian Groundwaters", National Symposium on Geogenic Contamination of Groundwater: Its Impact & Mitigation Strategies (GCG-2016), New Delhi, India (22 April, 2016). [Poster]
- **Ekamparam, A. S. S.**, and Pauline J., "Modeling of Transport and Bio-transformation of As (V) in Confined Aquifers", National Conference on Recent Innovations in Civil Engineering (RICE'14), Alagappa Chettiar College of Engg. & Tech., Karaikudi, India (28th May, 2014). [Oral]
- Nirmala, V., Leelavathy, K. R. and **Ekamparam, A. S. S.**, "A Fuzzy Approach to Find the Deficit Level of Dissolved Oxygen in Streams", National Conference on Applications of Basic Sciences in Engineering "NCABSE'11", S. A. Engineering College, Chennai, India (4 March, 2011) [Oral]

FUNDED PROJECTS

- Smart Groundwater Recharge: A Trans-disciplinary USA-India Framework for Eco-Resilient Groundwater Sustainability, Flood Mitigation, and Irrigation Security - Competitive Grants, OSU 2025-2026
  Supervisor: Dr. Salini Sasidharan; Oregon State University, USA
  Co-Investigator: Dr. Aravinth Ekamparam Budget: USD 50,000
- Optimizing Water Treatment and Soil Health in Agrivoltaic Systems for Sustainable Agriculture and Managed Aquifer Recharge in Oregon Competitive Agricultural Research Foundation Grants, 2025 - 2027
  Supervisor: Dr. Salini Sasidharan; Oregon State University, USA
  Co-Investigator: Dr. Aravinth Ekamparam Budget: USD 15,000

SUPERVISION/ MENTORSHIP EXPERIENCE

Oregon State University, 2022-Present

- Kanda, Taisuke (Dec 2024 - Oct 2025): Basics of performing column experiments, testing the efficiency of natural polymer on nitrate removal
- Spooner, Luke (Dec 2024 - Oct 2025): Installation of rainwater harvesting system to existing agrivoltaics infrastructure at North Willamette Research and Extension Center, Aurora, Oregon
- Goen L, Jackson (Dec 2024 - Oct 2025): Installation of rainwater harvesting system to existing agrivoltaics infrastructure at North Willamette Research and Extension Center, Aurora, Oregon
- Tomiatti Moura E Silva, Thomaz (Feb 2023 - June 2025): Establishing laboratory basic analysis protocol, Soil processing for particle size analysis, Determination of flow rate and mitigation for in situ clogging for an infiltration basin
- Namyasenko, Grigory (June 2024 - June 2025): Determination of flow rate and mitigation for in situ clogging for an infiltration basin
- Hill, Kaitlyn Mary (Mar 2023 - Sep 2023): Establishing laboratory analysis protocols for nitrate and turbidity, Soil processing for particle size analysis

TEACHING EXPERIENCE

Oregon State University, 2022-Present

- Assisted and taught field groundwater sampling and analysis procedure for graduate students at our field site at NWREC, Oregon (one day)

Indian Institute of Technology Madras, 2022

- Introduction to Visual MINTEQ for water chemistry
  2 Guest Lectures in Prof. Ligy Philip's course

National Institute of Technology Tiruchirappalli, 2021-2022

- Handled Strength of Materials Laboratory for undergraduate sophomore students
  Course code: CELR11; No. of students: 60; Teacher evaluation result: scored 3.9/5
- Taught Basics of Civil Engineering for undergraduate first year students
  Course code: CEIR11; No. of students: 60; Teacher evaluation result: scored 4.9/5
- Taught Environmental Process Chemistry and Microbiology for first year graduate students
  Course code: CE701; No. of students: 22; Teacher evaluation result: scored 3.75/5
- Handled Environmental Quality Measurements Laboratory for first year graduate students
  Course code: CE709; No. of students: 20; Teacher evaluation result: scored 3.96/5
- Taught Air Pollution and Control Engineering for first year graduate students
  Course code: CE706; No. of students: 20; Teacher evaluation result: scored 5/5
- Taught Industrial Wastewater Management for first year graduate students
  Course code: CE713; No. of students: 15; Teacher evaluation result: scored 5/5

Indian Institute of Technology Kanpur, 2014-2021

- Student tutor for Technical Arts (TA101; undergraduate core course)
  Assisted Prof.PM Mohite (Instructor) in teaching and evaluated student projects
- Teaching assistant for Surface Water Quality Modeling (CE760A; graduate modular course)
  Assisted Prof.Purnendu Bose (Instructor) in teaching and evaluated assignments, quiz copies, and student projects
- Teaching assistant for Solid-Water Interfacial Processes (CE767B; graduate modular course)
  Assisted Prof.Abhas Singh (Instructor) in teaching and evaluated assignments, quiz copies, and student projects
- Teaching assistant for Technical Arts (TA101; undergraduate core course)
  Assisted Prof.Mukesh Sharma (Instructor) in teaching and graded weekly course assignments

Alagappa Chettiar College of Engineering & Technology, Karaikudi, 2012-2014

- Teaching assistant for Environmental & Irrigation Drawing (undergraduate core course)
  Gave one lecture and Assisted Prof.Ezhil Priya (Instructor) in teaching, evaluated assignments, quiz copies, and student projects

INSTRUMENT EXPERIENCE

Oregon State University, 2022-Present

- Analyze water samples (various matrices) using inductively coupled plasma mass spectrometry (ICP-MS; Thermo iCAP RQ) and inductively coupled plasma optical emission spectrometry (ICP-OES)
- Analyze aqueous samples for nitrate, ammonium, and phosphate concentrations using the SEAL AQ300 instrument
- Measure particle size (submicron/micron range) using a particle size analyzer
- Analyze particle size of various nanoparticles and measure zeta potential in different matrices
- Operate pressure, turbidity, and water level sensors with dataloggers

Indian Institute of Technology Kanpur, 2014-2021

- Instrument In-charge (Under Prof. Vinod Tare): Maintained and analyzed water samples (various matrices) using Inductively Coupled Plasma Mass Spectrometry (ICP-MS) and Microwave Plasma Atomic Emission Spectrometry (MP-AES) (2 semesters)
- Instrument In-charge (Under Prof. Abhas Singh): Maintained and analyzed water samples (various matrices) using ICP-MS; developed methods for chromium and arsenic speciation and conducted analysis using ion chromatography coupled with ICP-MS (3 semesters)
- DLS Instrument – Experienced User, Environmental Geochemistry Lab (Under Prof.Abhas Singh): Analyzed particle size of various nano- and submicron particles; measured zeta potential across different matrices; conducted molecular weight measurements for internal and external institutional samples

TECHNICAL SKILLS

- Civil Engineering Application Software: ANSYS and STAAD
- Computational Fluid Dynamics Software: Fluent
- Environmental Application Software: Visual MINTEQ, Mineql+, Geochemists Work Bench
- Solid Phase Characterization Techniques:
  - Rietveld quantitative analysis on XRD patterns using MAUD and JADE software
  - Semi-quantitative elemental analysis on XPS spectra using CasaXPS software
  - HRTEM-SAED image analysis using GATAN software
- Computer Aided Drafting Software: AutoCAD
- Graphing Software: Origin
- Programming Software: Matlab (Basic), Python (Basic), R (Beginner)
- Productivity Software (Proficient): MS Word, MS Excel, MS PowerPoint

AWARDS AND HONORS

Oregon State University

- 2024: OSU Postdoctoral Association Professional Development Award, $500

Indian Institute of Technology Kanpur

- 2018: Goldschmidt Student Early Career Grant, $520
- 2014-2020: Institute Fellowship

Alagappa Chettiar Government College of Engineering, Tamil Nadu

- 2014: Overall, Third rank in the Environmental Engineering Specialization
- 2013: Government of India Scholarship, *Rs.* 15,000
- 2012-2014: TEQIP Scholarship, *Rs.* 1,32,000

University College of Engineering Tindivanam, Tamil Nadu

- 2012: Overall, Third rank in the Department of Civil Engineering
- 2009: First rank in the Department of Civil Engineering
- 2009-2012: Government of India Scholarship, *Rs.* 30,000

Government Higher Secondary School, Bhuthapandy, Tamil Nadu

- 2008: 12^{th} grade – Second rank – School level
- 2006: 10^{th} grade – First rank – School level
- 2004-2008: Tamil Nadu Government Talenter's Scholarship, *Rs.* 4000
- 2004-2005: Noorul Islam University Student Scholarship, *Rs.* 500
- 2003-2004: Noorul Islam University Student Scholarship, *Rs.* 500

UNFUNDED PROPOSALS

- Navigating Climate Uncertainty: The Role of Isotope Hydrology and Managed Aquifer Recharge in Water Sustainability – Northwest Climate Adaptation Science Center Postdoctoral Fellowship 2025-2026
  Supervisor: Dr. Salini Sasidharan; Oregon State University, USA
  Co-Investigator and Fellow: Dr. Aravinth Ekamparam; Proposed Budget:65,000
- Development of a Pretreatment Unit for Multiple Source Waters for Drywell Managed Aquifer Recharge (Drywell MAR) systems – WQRF Competitive Grants 2025
  Principal Investigator: Dr. Salini Sasidharan; Oregon State University, USA
  Co-Investigator: Dr. Aravinth Ekamparam; Proposed Budget:100,000
- Characterization of Fluoride Contaminated Soil Using Advanced Analytical Techniques for Mineral Identification – Client: Anchor QEA
  Principal Investigator: Dr. Salini Sasidharan; Oregon State University, USA
  Co-Investigator: Dr. Aravinth Ekamparam; Proposed Budget: USD 571,410
- Mechanistic Model for Competitive Binding of Chromium and NOM at Fe-oxide Surfaces
  HORIZON-MSCA-2023-PF-01 – Marie Skłodowska-Curie Actions
  Supervisor: Dr. Naresh Kumar; Wageningen University, The Netherlands
- Fulbright-Nehru Postdoctoral Fellowship 2021
  Supervisor: Dr. Jose M. Cerrato; The University of New Mexico, USA
- Chromium Mobilization by Manganese Carbonate
  HORIZON-MSCA-2021-PF-01 – Marie Skłodowska-Curie Actions
  Supervisor: Dr. Naresh Kumar; Wageningen University, The Netherlands

Professional Service

Editorial Board Member

- American Journal of Environmental Science and Engineering

Reviewer for

- ACS Omega

- Groundwater for Sustainable Development
- Clays and Clay Minerals
- Chinese Journal of Geochemistry
- Water SA

Professional Affiliations

- American Chemical Society
- Geochemical Society
- American Geophysical Union

Volunteering Activities

- Vice President - OSU Postdoctoral Association (Apr 2025 - Oct 2025)
- Guest Room Secretary - Hall VII, IIT Kanpur (Apr 2017 - Apr 2018)

REFERENCES

- Dr. Salini Sasidharan - Assistant Professor, Department of Biological and Ecological Engineering, Oregon State University, Corvallis, Oregon, United States - 97331,
email: salini.sasidharan@oregonstate.edu, Mobile: +1 (310) 216 8836, (Postdoc Supervisor).
- Dr. Scott Bradford – Research Leader, ARS, Sustainable Agricultural Water Systems Unit, U.S. Department of Agriculture, 239 Hopkins Road, Davis, CA, United States - 95616,
email: scott.bradford@usda.gov, Office: +1 (530) 754 3351, (Postdoc Co-supervisor).
- Dr. Abhas Singh - Professor, Department of Civil Engineering, Indian Institute of Technology Kanpur (IIT Kanpur), Kanpur Uttar Pradesh, India 208 016,
email: abhas@iitk.ac.in, Office: +91 (512) 259-7665 (PhD Advisor).
- Dr. S. Pauline - Associate Professor, Department of Civil Engineering, Government College of Engineering, Tirunelveli, Tamil Nadu, India 627 007, email: pauline@gcetly.ac.in, Mobile: +91 (770) 859-3001 (Masters Thesis Advisor).
- Dr. Sumant Kumar – Scientist E, Ground Water Hydrology Division, National Institute of Hydrology, Roorkee, Uttarakhand, India 247667, email: sumantk.nihr@gov.in, Mobile: +91 (925) 803-3620 (Collaborator).